Q1) Consider the simple Keynesian Multiplier model with con sumption function given by Ct = a + bYt−1 and with con stant investment and government expenditures. Solve and simulate the model. Plot the time series of output (Yt) values for different values of the MPC parameter

Given Keynesian multiplier model:

We are in a Keynesian multiplier model with lagged consumption:

$C_t=a+bY_{t-1}$, $0<b<1$

Government spending and investment are constant: (given)

$I_t=I^-$, $G_t=G^-I$

So equilibrium income
$Y_t = C_t+I_t+G_t$

The we define $A = a +I^- +G^-$
Then $Y_t=A+bY_{t-1}$

Solving the difference equation:
$Y_t=A+bY_{t-1}$ where $A:=a+I^-+G^-$, $0<b<1$

$Y_t - bY_{t-1} = A$

Homogenous solution :

$Y_t^{(h)} - bY_{t-1}^{(h)} = 0$
$Y_t^{(h)} = bY_{t-1}^{(h)}$
Doing this step by step

$Y_1^{(h)} = bY_0^{(h)}$
$Y_2^{(h)} =bY_1^{(h)} = b(bY_0^{(h)})) = b^2Y_0^{(h)}$
$Y_3^{(h)} =bY_2^{(h)}= b(b2Y0(h))=b^3Y_0^{(h)}$ …etc

By continuing this multiplication t times we get

$Y_t^{(h)} = b^t Y_0^{(h)}$
Writing the $Y_0^{(h)}$ as constant C determined by the initial condition we get
$Y_t^{(h)} = Cb^t$ _____Homogenous solution

Now, solving for particular solution,
(Because RHS is a constant A, we find a constant particular solution $Y_t^{(p)}=Y_p$

Then,
$Y_p = A + bY_p$
$Y_p - bY_p = A$ (Assuming 1-b is not equal to 0)
$(1-b)\ Y_p = A$
$Y_p = A/(1-b)$ (note here since the RHS was constant out particular solution is independent of time)

$Y^* = A/(1-b)$ _____Particular solution

Now, solving for the general solution

$Y_t = Y_t^{(p)} + Y_t^{(h)} = A/(1-b) + Cb^t$ _____General solution

Determining C

If the initial value Y_0 is given (at t=0), plugging t=0:

$Y_0 = A/(1-b) + Cb^0 = A/(1-b) + C$
$C = Y_0 - A/(1-b)$

So then, general solution :

$Y_t = A/(1-b) + (Y_0 - A/(1-b))b^t$
$Y_t = (Y_0 - Y^*)b^t + Y^*$

Why do we assume 0<b<1 ?
Mathematical reasoning:

We had the solution

$Y_t = (Y_0 - Y^*)b^t + Y^*$
$Y^* = A/(1-b)$

- The term b^t governs the dynamics.
    - If $|b| < 1$, then b_t tends to 0 as t tends to infinity. Thus Yt tends to Y*. The economy converges to a stable equilibrium.
    - If $|b| \geq 1|$, the multiplier dynamics do not converge:
        - b = 1: $Yt = Y_0 + At$ i.e. income grows without bound.
        - b>1b: b^t diverges, so income explodes.

    - $b \leq -1$: path oscillates violently or diverges.

- The steady state itself,
  $Y^* = A/(1-b)$
  only makes sense if $1-b > 0$ i.e. $b<1$. Otherwise the denominator is 0 or negative, giving infinite or unstable values.

Economic reason:

If b=0: households consume nothing out of additional income. In reality, people do spend some fraction of new income.

If b=1 households consume all of any additional income, leaving nothing for saving. Then saving = 0 at every income, which contradicts the need for investment financing in a Keynesian framework.

If b>1: households spend more than their income increase, i.e. dissave every time income rises unsustainable in the long run.

If b<0: households reduce consumption when income rises - unrealistic for normal consumption behavior.

Stability conditions:

The general solution of the model is

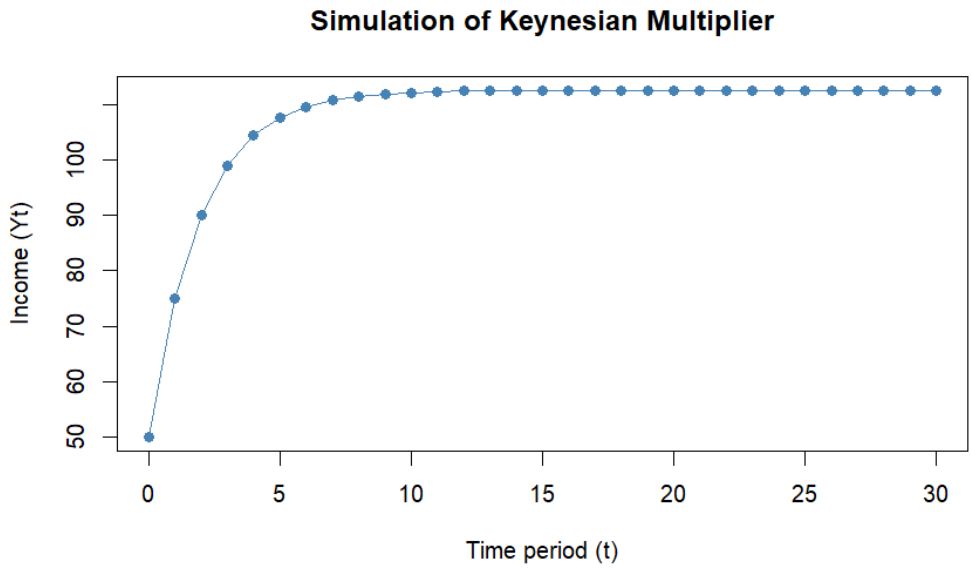
$Y_t = (Y_0 - Y^*)b^t + Y^*$

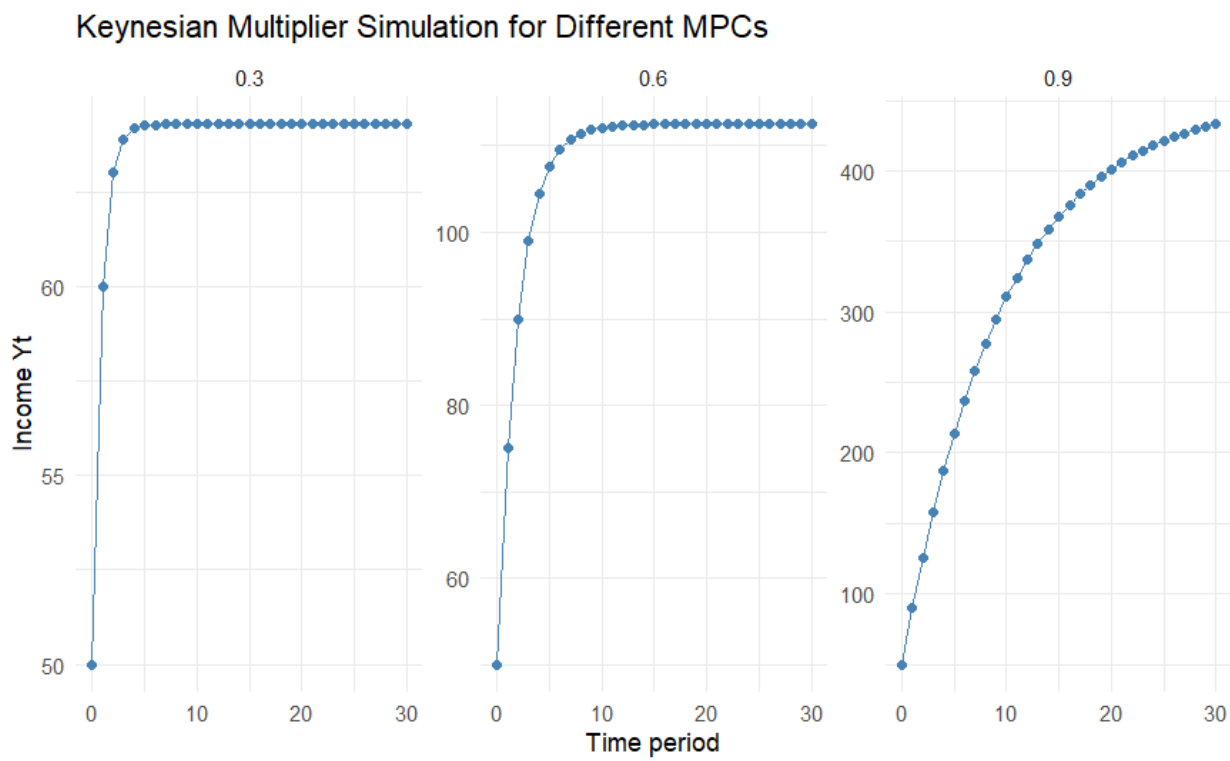
where $Y^* = A/(1-b)$ is the steady state. For Y_t to converge to Y^* as t tends to infinity, the term b^t must go to zero. This requires $|b| < 1$

If $|b| >= 1$, the deviations from the steady state either do not decay (b=1) or diverge ($|b| > 1$), making the system unstable.

Economically, b is the marginal propensity to consume. When $0< b <1$, each additional unit of income is partially consumed, so the economy gradually converges to the steady-state income Y^*. If $-1 < b < 0$, the system is still stable, but the path of income oscillates around the steady state, alternating above and below Y^* before settling.

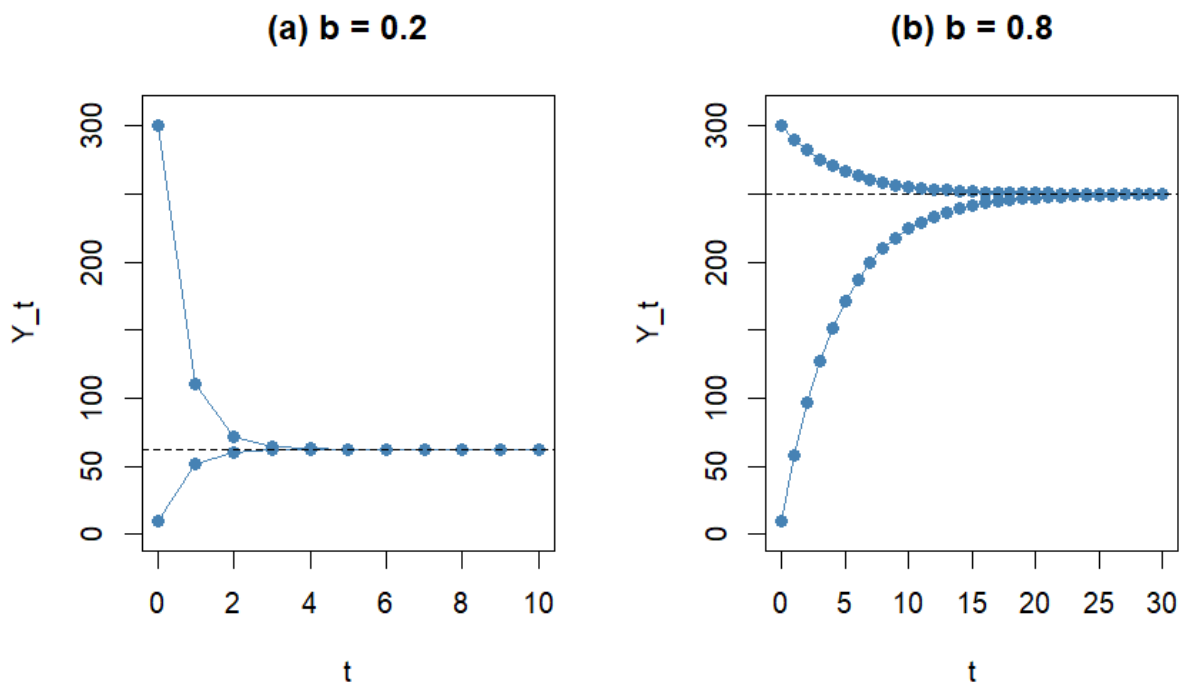
Unstable cases : b=1- income grows linearly without bound. : b>1 - income diverges exponentially. b <= -1 - income oscillates with increasing amplitude and never reaches the steady state





The graph shows the time path of income Y_t for different values of the marginal propensity to consume (MPC). For all cases, income starts at the initial value Y_0 and gradually converges to the steady-state level Y* = A/ (1-b). As the MPC increases, the rate of convergence slows, higher MPC values (closer to 1) cause income to move more gradually toward the steady state, reflecting stronger multiplier effects and larger responses to previous-period income.

Conversely, lower MPC values lead to faster convergence, as smaller consumption responses dampen the dynamics. The figure clearly illustrates that the speed at which the economy approaches equilibrium is inversely related to the MPC.



The graphs show the evolution of output Y_t under the Keynesian multiplier model for two different values of the marginal propensity to consume. For b=0.2 (plot a), the equilibrium output is relatively low, and output quickly converges to this level. For b=0.8 (plot b), the equilibrium is much higher, and convergence is slower, reflecting a stronger multiplier effect. The different initial values mainly affect the early trajectory, but over time each path approaches its respective equilibrium. Overall, higher b increases the equilibrium income and slows the adjustment process.

Q2) Solve and determine the solution path of Yt.

Given

$Y_t=C_t+I_t$,
$C_t,=a+bY_{t-1}$,
$I_t=hY_{t-1}$

Substituting C_t I_t into Y_t:

$Y_t = a + bY_{t-1}+ hY_{t-1}= a+(b+h)\, Y_{t-1}$.

Let ρ:=b+h The model is the first-order linear difference equation

$Yt= a+\rho Y_{t-1}$ with ρ=b+h.

Consider the equation $Yt^{(h)} = \rho\, Y_{t-1}^{(h)}$ by setting a = 0
Trying solution $Y_t^{(h)}=\lambda^t$, substituting,
$\lambda^t = \rho\lambda^{t-1}$
Dividing both sides by λ^{t-1} (assuming λ is not equal to 0)
λ=ρ

So λ=ρ is the characteristic root. Therefore any multiple of ρ^t is a solution:

$Y_t^{(h)} = C\rho^t$ —---------- Homogeneous solution

where C is an arbitrary constant determined by the initial condition.
(the λ=0 case is trivial if ρ=0 then $Y_t^{(h)}= 0$)

Now, consider to get the particular solution, trying $Y_t^{(p)} =Y^*$
$Y^* = a +\rho Y^*= (1-\rho)Y^*= a$

If ρ not equal to 1

$Y^* = a/1-\rho$ —---------- Particular solution

So then the general solution,
$Y_t =Y^*+C\rho^t = a/1-\rho + C\rho^t$ —----------General solution

By iteration,

$Y_1 = a + \rho\, Y_0$
$Y_2 = a + \rho\, Y_1 = a + \rho\,(a + \rho\, Y_0) = a\,(1 + \rho) + \rho^2\, Y_0$
⋮
$Y_t = \rho^t Y_0 + a\sum\{k=0\}^\{t-1\} \rho^k = \rho^t Y_0 + a(1-\rho^t)/(1-\rho)$, (ρ not equal to 1)

$\rho^t Y_0 + a(1-\rho^t)/(1-\rho) = a/(1-\rho) + \rho^t [Y_0 - a/(1-\rho)]$

Stability conditions

Steady:

A steady state Y^* satisfies
$Y^* = a+\rho Y$

Solving for Y^* gives
$Y^*=a/1-\rho$, valid if ρ not equal to 1

Deviations from the steady state

Define the deviation from equilibrium:
$x_t=Y_t-Y^* x_t=Y_t-Y^*$

Substitute and get:
$X_t = Y_t-Y^* = \rho^t(Y_0-Y^*)$

So then dynamics would be through
$X_t = \rho x_{t-1}$

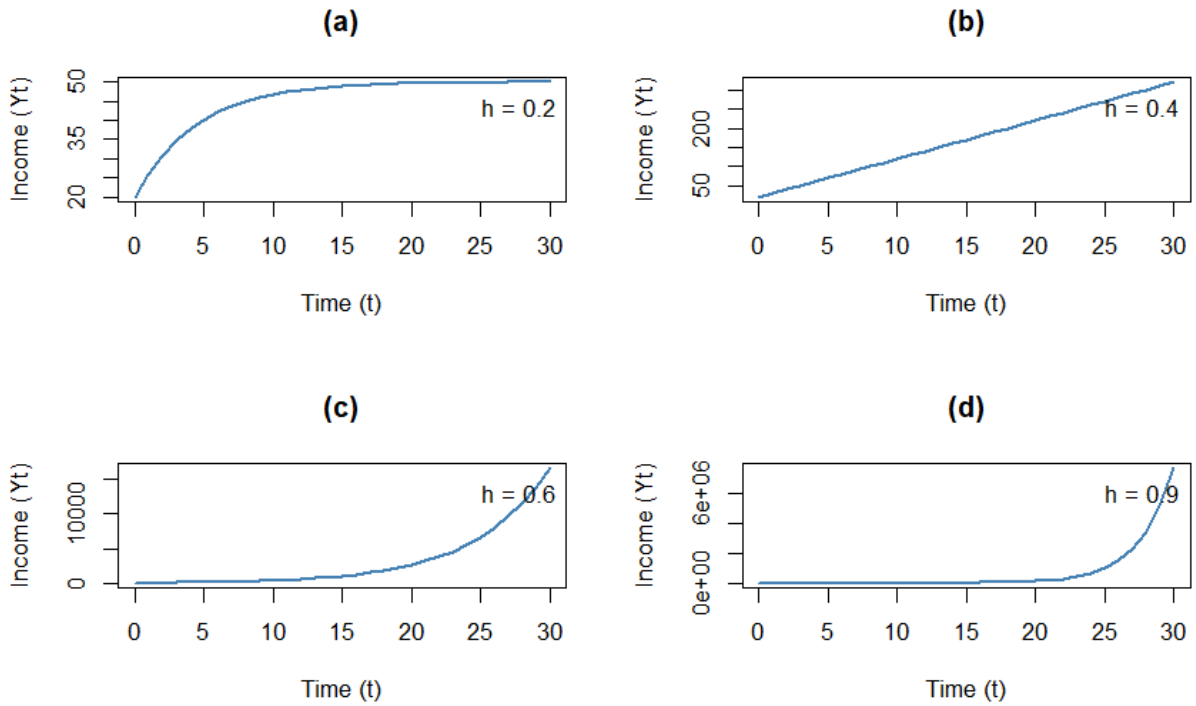
For stability, we require that deviations x_t vanish as t tends to infinity
That happens only if $|\rho| < 1 == |b+h| < 1$

Economic intuition: If $0<= b<1$ (marginal propensity to consume) and $h>=0$ (investment responsiveness), then $0<= b+h < 1$
This ensures the system converges to the steady-state income Y^*

If instead:

- $b+h > 1$: income grows without bound (explosive).
- $b+h=1$: income grows linearly with time (no finite equilibrium).
- $-1<b+h<0$: converges to equilibrium but with oscillations (up-down pattern)

The keynesian multiplier is stable iff $|b+h| <1$



(a) h = 0.2 & ρ = 0.8 (stable case)

- Steady state: Y* = 10 / (1 – 0.8) = 50
- By t = 30: $Y_{30} \approx 49.96$
- Behavior: Yt rises smoothly from the initial value (20) and converges to the steady state 50. The convergence is monotonic and stable.
- ρ=0.8 has magnitude less than 1, so ρt tends to 0 and the deviation term ρt(Y0−Y*) vanishes. Economically this means induced spending (MPC b plus investment-response h) is not large enough to produce run-away feedback; the multiplier 1/1−ρ=5 is finite.

(b) h = 0.4 & ρ = 1.0 (unit root case)

- No finite steady state.
- The recurrence is $Y_t = Y_0 + at$
- By t = 30: $Y_{30} = 20 + 10\times30 = 320$
- Behavior: Yt grows linearly without bound. Each period, income increases by exactly 10 units.
- With ρ=1 each period autonomous spending a is simply added on top of last period's income with no damping. Economically this is the knife-edge case: multiplier is infinite in the steady-state formula, and shocks produce permanent, linear drift.

(c) h = 0.6 & ρ = 1.2 (explosive case)

- Algebraic steady state: $Y^* = 10 / (1 – 1.2) = –50$ (exists mathematically but is unstable).
- By t = 30: $Y_{30} \approx 16{,}582$

- Behavior: Yt grows exponentially, diverging away from the unstable fixed point. The growth accelerates sharply after a few periods.
- $\rho>1$ means the deviation term $\rho^t(Y_0-Y^*)$ blows up in magnitude. Although algebraically there is a fixed point at $Y^*=-50$, that fixed point is **unstable**: any realistic initial condition (here $Y_0=20$) drives Y_t away from it. Economically, induced responses are strong enough that demand feedback feeds on itself and becomes explosive.

(d) h = 0.9 & ρ = 1.5 (strongly explosive case)

- Algebraic steady state: $Y^* = 10 / (1 – 1.5) = –20$ (unstable).
- By t = 30: $Y_{30} \approx 7{,}700{,}000$
- Behavior: Very rapid exponential growth. The trajectory appears flat at first but then shoots upward sharply, reflecting runaway instability.
- Higher h increases ρ and hence the growth rate ρ^t. The larger ρ, the faster the exponential explosion. This is the dramatic “runaway multiplier” case: a small increase in h (from 0.6 to 0.9 here) greatly magnifies the outcome because exponentials accelerate rapidly.

Stability requires $|\rho| <1$, i.e. $b+h<1$ (since here all parameters are positive). If the sum of the MPC and the investment-responsiveness is below 1, the economy converges to a finite steady income after shocks.

At $\rho=1$ we get persistent, un-damped accumulation of the autonomous component (unit root).

For $\rho>1$ the system is unstable and income explodes (or diverges, possibly with oscillation if $\rho<0$). Economically this means the feedback from current income to next period’s induced spending is too strong, policymakers should be wary of mechanisms that push b+h above 1, since small shocks get amplified without bound here.

Q3) Assuming na¨ıve expectations for the expected disposable income YDe, analytically derive the equilibrium level of income and reproduce Table 3.6 on Page 81 of the Godley and Lavoie book.

Steady state (denoting steady values with no time subscript). In steady state expectations are fulfilled so $Y_D^e = Y_D = (1-\theta)Y$. Also $\Delta H^s = 0$, $G-\theta Y = 0$

From $\Delta H^s = 0$ we get the steady income:

$\Delta H^s = G-T$,

$\Delta H^s = 0$

$0 = G-T$: $T=G$

$T=\theta Y$

$\theta Y = G$ so,

$Y^* = G/\theta$

Using $C=\alpha_1(1-\theta)Y + \alpha_2 HC$ and the steady-state identifying $C=Y_D=(1-\theta)Y$, equating:

$(1-\theta)Y=\alpha_1(1-\theta)Y + \alpha_2 H$

$(1-\alpha 1)(1-\theta)Y = \alpha_2 H$, $H^*= (1-\alpha 1)(1-\theta)/\alpha_2(Y^*)$

So then consumption in steady state is : $C^* = (1-\theta)Y^*$

which equivalently equals $\alpha_1(1-\theta)Y^*+\alpha_2 H^*$ once H^* is substituted).

So,

$Y^*=G/\theta$,

$H^*=(1-\alpha_1)(1-\theta)/\alpha_2 (Y^*)$

Parameters:

- $\alpha 1=0.6$, $\alpha 2=0.4$
- $\theta=0.2$ (so tax rate 20%)
- initial (pre-shock, period 1 baseline) household money stock $H_{-1}=0$ and the economy is initially at the pre-shock level $Y_{-1}=20$, hence $YD_{-1}=(1-\theta)Y_{-1}=16$
- Government path: G1=0, G2=20, G3=20, (i.e. a $20 increase in government expenditure starting in period 2).
- Naïve expectations: $Y_{D,t}^e = Y_{D,t-1}$

Then,

$Y^* = G/\theta = 20/0.2 = 100$, $YD^* = (1-\theta)Y^* = 0.8\times100 = 80$

Taking $\alpha_1=0.6$, $\alpha_2=0.4$

$H^*=(1-0.6)(1-0.2)/0.4\ Y^*=0.4\times0.80/0.4\ (Y^*) =0.8\ Y^*=80$

Period by period, reproducing the table

Period 1 = pre-shock (G1 =0). We take the economy to have been at a level such that $YD_1=16$(that is the lagged disposable income used to form naive expectations for period 2).

Period 2: government raises G to 20; expectations are naive so $Y_{D,2}^e=YD_1=16Y$. Then finding C2, Y2, T2, ΔHC etc.

Period 3: expectations $Y_{D,3}^e= Y_{D2}$ and so on

Replicating the table in excel:

Period	1.0	2.0	3.0	4.0	5.0	6.0	7.0	8.0	9.0	10.0
$YD_t^e = YD_{t-1}$	0.0	16.0	23.7	31.9	39.7	46.6	52.6	57.7	61.9	65.4
C_t	0.0	9.6	19.8	29.6	38.3	45.8	52.1	57.4	61.7	65.3
$Y_t = C_t + G_t$	0.0	29.6	39.8	49.6	58.3	65.8	72.1	77.4	81.7	85.3
T_t	0.0	5.9	8.0	9.9	11.7	13.2	14.4	15.5	16.3	17.1
$YD_t = Y_t - T_t$	0.0	23.7	31.9	39.7	46.6	52.6	57.7	61.9	65.4	68.2
$deltaHs_t = G_t - T_t$	0.0	14.1	12.0	10.1	8.3	6.8	5.6	4.5	3.7	2.9
$deltaHh_t = YD_t - C_t$	0.0	14.1	12.0	10.1	8.3	6.8	5.6	4.5	3.7	2.9
$H_t = H_{t-1} + deltaHs_t$	0.0	14.1	26.1	36.2	44.5	51.4	57.0	61.5	65.1	68.1
$deltaHd_t = YD_t^e - C_t$	0.0	6.4	3.8	2.3	1.4	0.8	0.5	0.3	0.2	0.1
$Hd_t = Hd_{t-1} + deltaHd_t$	0.0	6.4	17.9	28.4	37.6	45.4	51.9	57.3	61.7	65.2
G	0.0	20.0	20.0	20.0	20.0	20.0	20.0	20.0	20.0	20.0

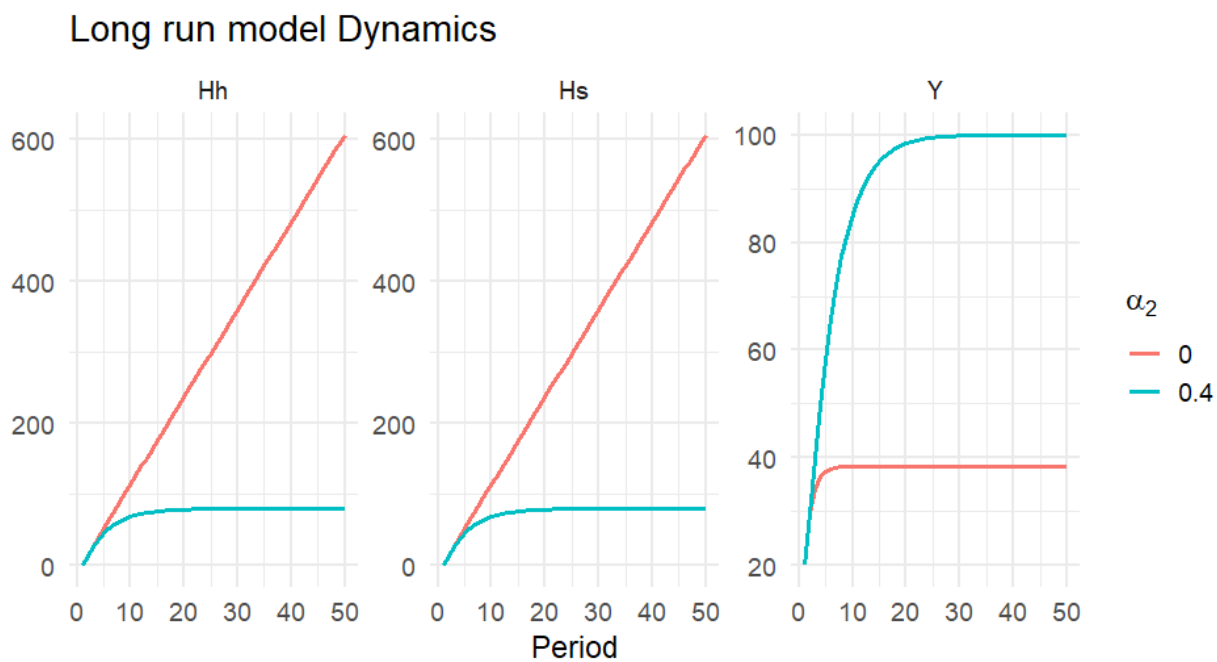
This when done till infinity converges as follows:

Period	33.0	34.0	35.0	36.0	37.0
$YD_t^e = YD_{t-1}$	79.9	79.9	79.9	80.0	80.0
C_t	79.9	79.9	79.9	80.0	80.0
$Y_t = C_t + G_t$	99.9	99.9	99.9	100.0	100.0
T_t	20.0	20.0	20.0	20.0	20.0
$YD_t = Y_t - T_t$	79.9	79.9	80.0	80.0	80.0
$deltaHs_t = G_t - T_t$	0.0	0.0	0.0	0.0	0.0
$deltaHh_t = YD_t - C_t$	0.0	0.0	0.0	0.0	0.0
$H_t = H_{t-1} + deltaHs_t$	79.9	79.9	80.0	80.0	80.0
$deltaHd_t = YD_t^e - C_t$	0.0	0.0	0.0	0.0	0.0
$Hd_t = Hd_{t-1} + deltaHd_t$	79.9	79.9	79.9	80.0	80.0
G	20.0	20.0	20.0	20.0	20.0

Link to the spread sheet: [Michelle_assignment_3]

Plot the long run behavior of income, household wealth, government debt in this model for α2 = 0 and α2 = 0.4

Plotting with 0, 0.4 alpha values



This shows the importance of the parameter alpha2, which represents the marginal propensity to consume out of wealth. When α2=0, households do not consume out of their stock of wealth, but when α2=0.4, a fraction of the wealth stock affects consumption behavior.

- **For α2=0 (red line):**
  - Household wealth (Hh): Grows without bound, linearly over time.
  - Government debt (Hs): Also grows linearly and unbounded.
  - Income (Y): Stabilizes at a lower value, reflecting that the multiplier effect is only due to current income without any feedback from accumulated wealth.
- **For α2=0.4 (blue line):**
  - Household wealth (Hh): Stabilizes at a finite, steady-state level.
  - Government debt (Hs): Also stabilizes at a steady level.
  - Income (Y): Stabilizes at a much higher steady-state level compared to the scenario without wealth-induced consumption. This reflects a larger multiplier, where past wealth boosts ongoing consumption and output

Mathematically, when α2>0, household consumption depends not only on current disposable income expectations but also accumulates effects from past savings (wealth stock). This feedback mechanism causes both output and wealth to converge to steady-state values. In

contrast, with $\alpha_2=0$, there is no such stabilization, creating ever-increasing wealth and government debt but limited stabilization in output. This aligns with

- Steady-state income: $Y^*=G/\theta$
- Steady-state household wealth: $H^*=[(1-\alpha_1)(1-\theta)/\alpha_2]Y^*$

Economic Implications

- Inclusion of the wealth effect ($\alpha_2>0$) means fiscal expansion (increase in G) leads to higher equilibrium income and stable government debt, which is generally more sustainable for long-run economic growth.
- Excluding wealth effects ($\alpha_2=0$) means fiscal policy has weaker output effects, and the system generates ever-growing government liabilities.

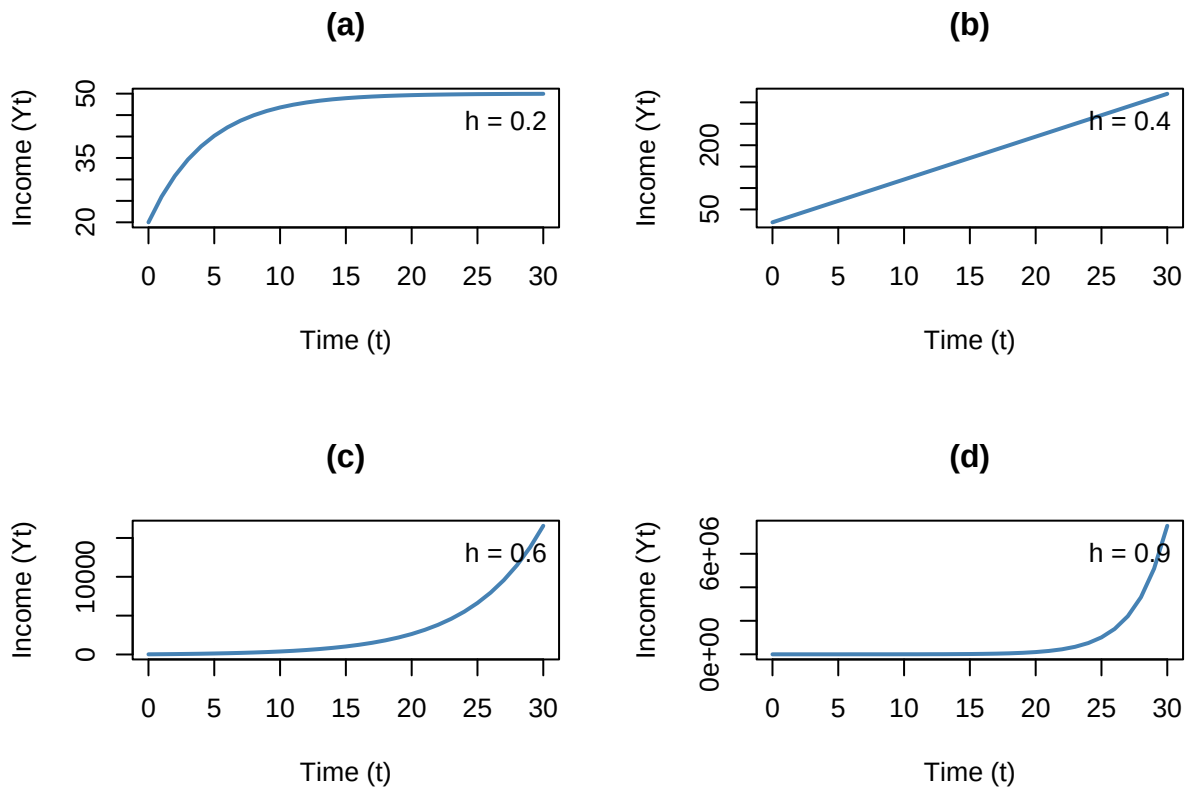
Scenario	Hh (Wealth)	Hs (Debt)	Y (Output)
$\alpha_2=0$	Linear growth	Linear growth	Stabilizes at low level
$\alpha_2=0.4$	Stabilizes	Stabilizes	Stabilizes at high level

Michelle_Assignment_3

Michelle

2025-09-28

```r
a <- 10       # autonomous expenditure
b <- 0.6      # MPC
Y0 <- 20      # initial income
t_max <- 30   # number of periods

# Function to simulate Yt
simulate_Y <- function(a, b, h, Y0, t_max) {
  Y <- numeric(t_max + 1)
  Y[1] <- Y0
  rho <- b + h
  for (t in 2:(t_max+1)) {
    Y[t] <- a + rho * Y[t-1]
  }
  return(Y)
}

# Choosing 4 different h values
h_values <- c(0.2, 0.4, 0.6, 0.9)

# Set up 2x2 plotting layout
par(mfrow = c(2,2))

# Looping over h values and plot
for (i in 1:4) {
  h <- h_values[i]
  Yt <- simulate_Y(a, b, h, Y0, t_max)
  plot(0:t_max, Yt, type="l", lwd=2,
       xlab="Time (t)", ylab="Income (Yt)",
       main=paste0("(", letters[i], ")"),
       col="steelblue")
  legend("topright", legend=paste("h =", h), bty="n")
}
```




```r
simulate_model <- function(alpha2, T_max = 30, alpha1 = 0.6, theta = 0.2, G = 20) {

  # Initialize vectors
  Y  <- numeric(T_max)
  T  <- numeric(T_max)
  YD <- numeric(T_max)
  YDe <- numeric(T_max)
  C  <- numeric(T_max)
  Hh <- numeric(T_max)    # household wealth
  Hs <- numeric(T_max)    # government debt

  # Initial conditions
  Y[1]  <- 20
  T[1]  <- theta * Y[1]
  YD[1] <- Y[1] - T[1]
  YDe[1] <- YD[1]
  C[1]  <- alpha1 * YDe[1] + alpha2 * 0
  Hh[1] <- 0
  Hs[1] <- 0

  # Simulation loop
  for (t in 2:T_max) {
    # Expectations: naive
    YDe[t] <- YD[t-1]

    # Consumption
```

```r
    C[t] <- alpha1 * YDe[t] + alpha2 * Hh[t-1]

    # Output
    Y[t] <- C[t] + G

    # Taxes
    T[t] <- theta * Y[t]

    # Disposable income
    YD[t] <- Y[t] - T[t]

    # Changes in wealth and debt
    dHh <- YD[t] - C[t]
    dHs <- G - T[t]

    # Update stocks
    Hh[t] <- Hh[t-1] + dHh
    Hs[t] <- Hs[t-1] + dHs
  }

  data.frame(
    period = 1:T_max,
    Y = Y,
    Hh = Hh,
    Hs = Hs,
    alpha2 = alpha2
  )
}

#running simulations
sim0   <- simulate_model(alpha2 = 0,   T_max = 50)
sim04  <- simulate_model(alpha2 = 0.4, T_max = 50)

sim_all <- rbind(sim0, sim04)

#reshaping the data to long form
library(dplyr)
```

```
## Warning: package 'dplyr' was built under R version 4.3.3

##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

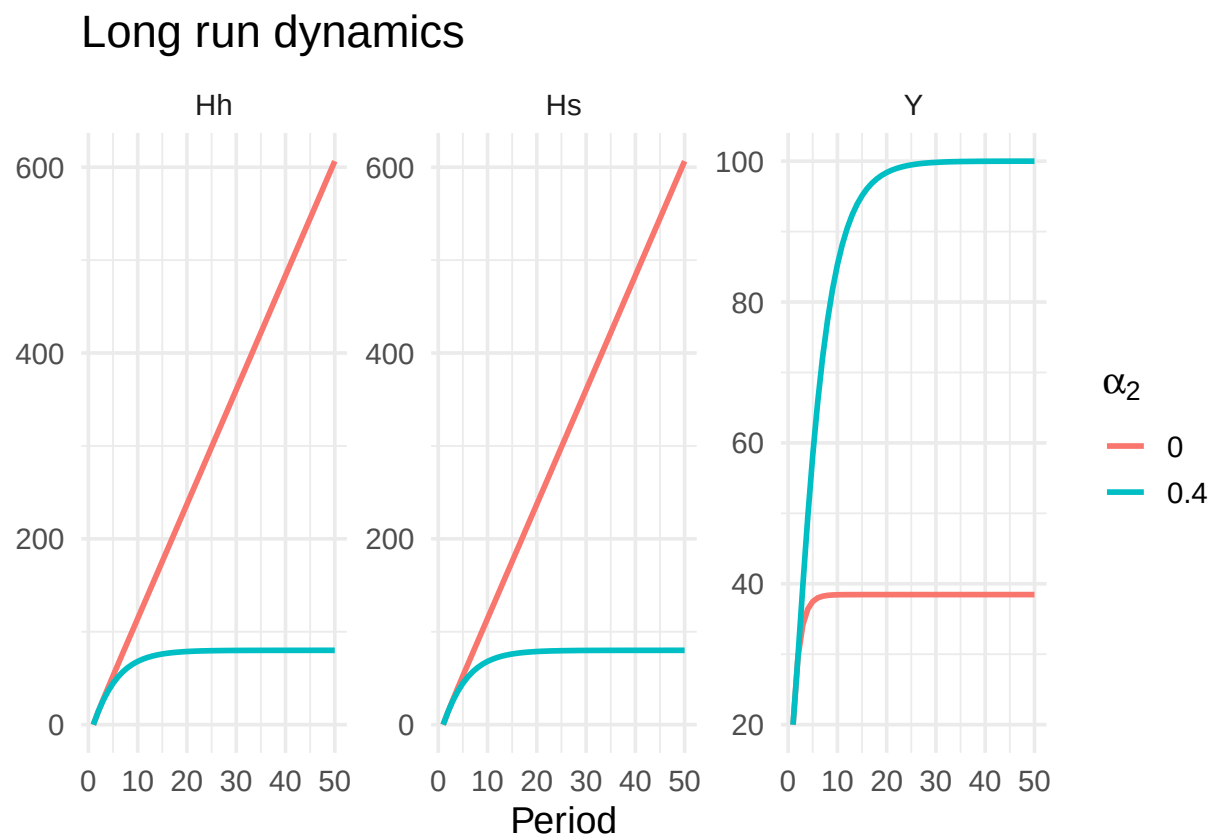
```r
library(tidyr)
```

```
## Warning: package 'tidyr' was built under R version 4.3.3
```

```r
library(ggplot2)

sim_long <- sim_all %>%
  pivot_longer(cols = c(Y, Hh, Hs),
               names_to = "variable",
               values_to = "value")


p_all <- ggplot(sim_long, aes(x = period, y = value, color = factor(alpha2))) +
  geom_line(size = 1) +
  facet_wrap(~ variable, scales = "free_y", ncol = 4, nrow = 4) +
  labs(title = "Long run dynamics",
       y = NULL,
       x = "Period",
       color = expression(alpha[2])) +
  theme_minimal(base_size = 14)
```

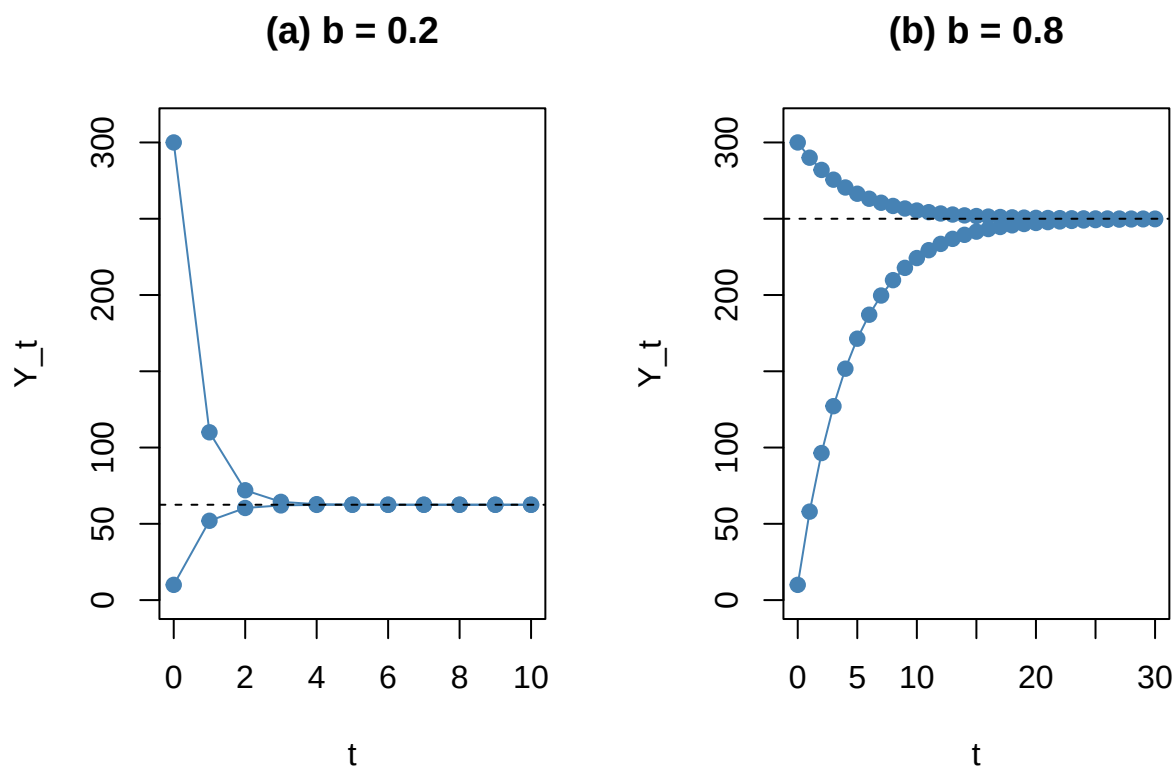
```
## Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use `linewidth` instead.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```

```r
print(p_all)
```



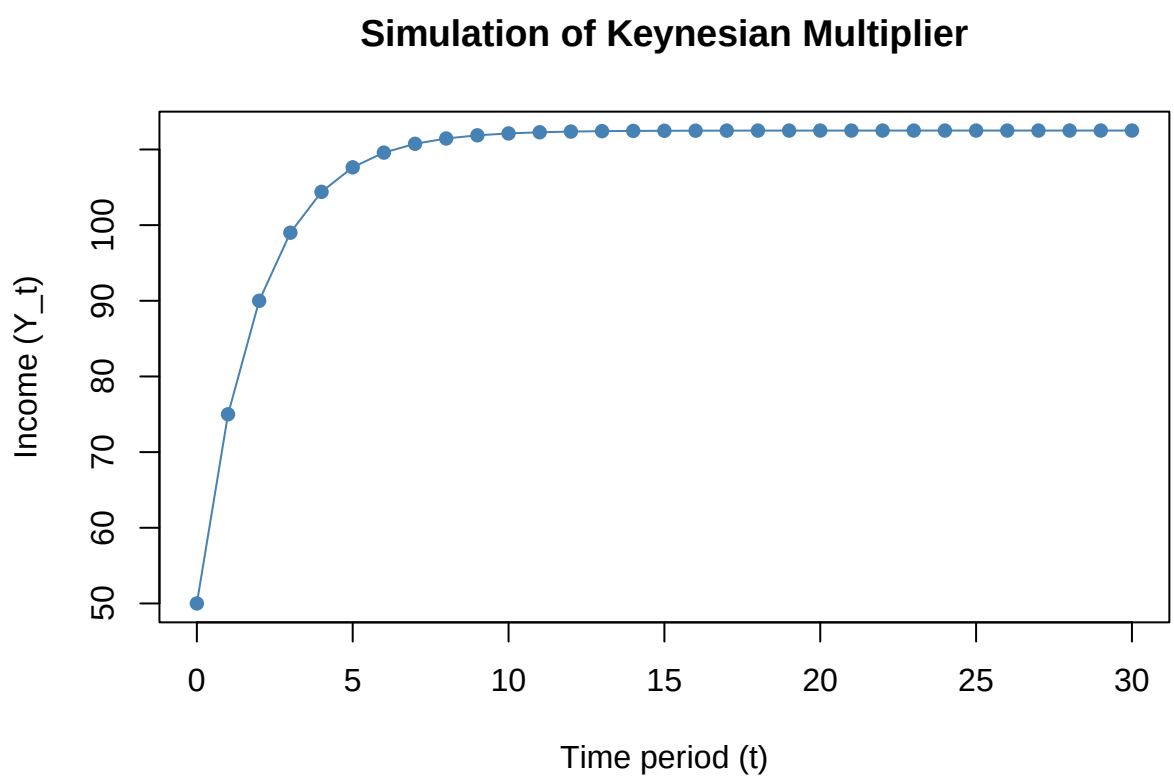

```r
a <- 10
I <- 20
G <- 20

t_max_a <- 10
t_max_b <- 30

# Function
simulate_Y <- function(Y0, b, a, I, G, t_max) {
  Y <- numeric(t_max + 1)
  Y[1] <- Y0
  for (t in 1:t_max) {
    Y[t + 1] <- a + b * Y[t] + I + G
  }
  return(Y)
}

# Part (a): b = 0.2
b_a <- 0.2
Y0_a <- c(10, 300)
Y_a1 <- simulate_Y(Y0_a[1], b_a, a, I, G, t_max_a)
Y_a2 <- simulate_Y(Y0_a[2], b_a, a, I, G, t_max_a)

# Part (b): b = 0.8
b_b <- 0.8
Y0_b <- c(10, 300)
```

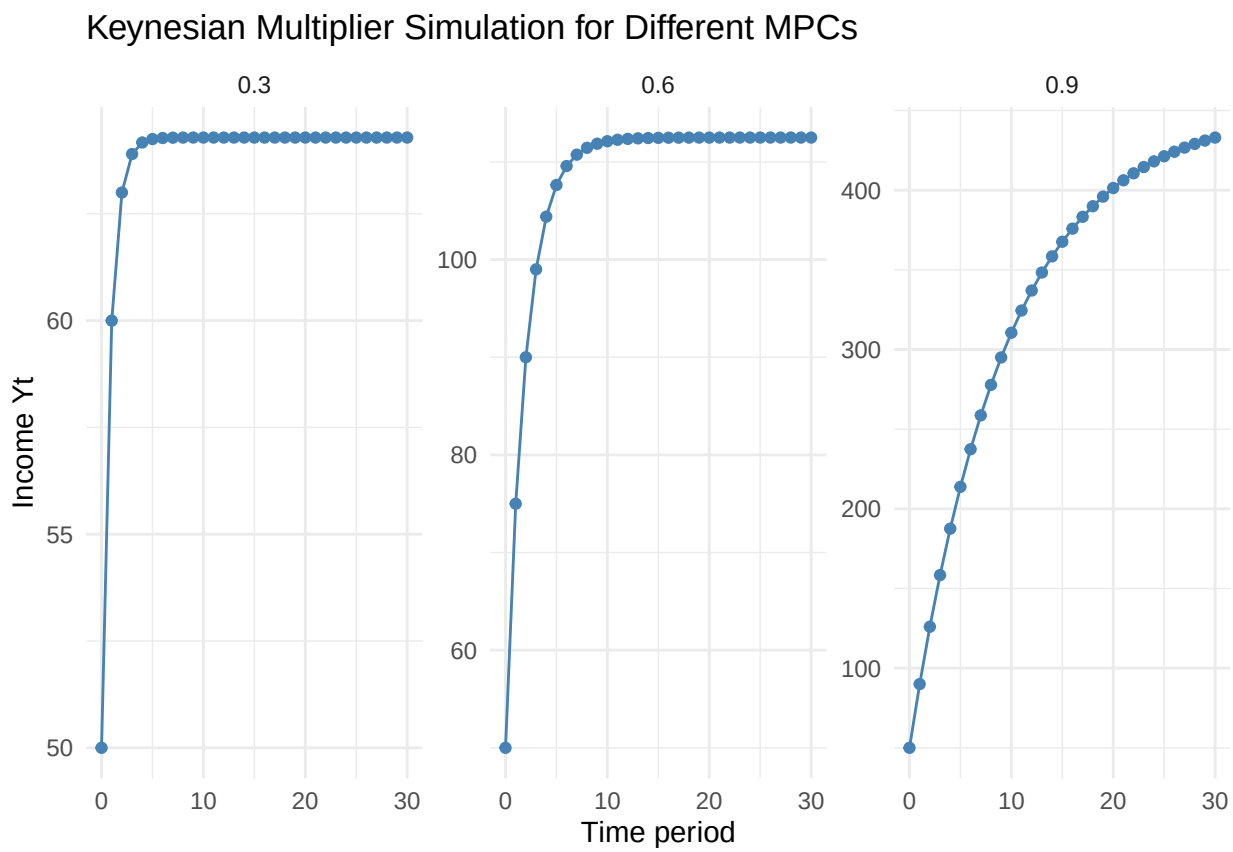
```r
Y_b1 <- simulate_Y(Y0_b[1], b_b, a, I, G, t_max_b)
Y_b2 <- simulate_Y(Y0_b[2], b_b, a, I, G, t_max_b)

# Plot
par(mfrow=c(1,2))  # side-by-side plots

# Plot (a)
plot(0:t_max_a, Y_a1, type='o', pch=19, col='steelblue', ylim=c(0,310),
     xlab='t', ylab='Y_t', main='(a) b = 0.2')
lines(0:t_max_a, Y_a2, type='o', pch=19, col='steelblue')
abline(h=(a + I + G)/(1-b_a), lty=2)  # equilibrium

# Plot (b)
plot(0:t_max_b, Y_b1, type='o', pch=19, col='steelblue', ylim=c(0,310),
     xlab='t', ylab='Y_t', main='(b) b = 0.8')
lines(0:t_max_b, Y_b2, type='o', pch=19, col='steelblue')
abline(h=(a + I + G)/(1-b_b), lty=2)  # equilibrium
```




```r
a  <- 10        # autonomous consumption
b  <- 0.6       # marginal propensity to consume
I  <- 20        # constant investment
G  <- 15        # constant government expenditure
A  <- a + I + G # constant term in recursion

Y0 <- 50        # initial income
```

```r
T  <- 30       # number of periods to simulate

# Initialize vector
Y <- numeric(T+1)
Y[1] <- Y0  # set initial income

# Iterative simulation
for (t in 1:T) {
  Y[t+1] <- A + b * Y[t]
}

# Time periods
time <- 0:T

# Plot the results
plot(time, Y, type="o", pch=16, col="steelblue",
     xlab="Time period (t)", ylab="Income (Y_t)",
     main="Simulation of Keynesian Multiplier")
```




```r
library(ggplot2)
b_values <- c(0.3, 0.6, 0.9)
sim_data <- data.frame()

for (b in b_values) {
  Y <- numeric(T+1)
```

```r
  Y[1] <- Y0
  for (t in 1:T) {
    Y[t+1] <- A + b*Y[t]
  }
  sim_data <- rbind(sim_data, data.frame(Time=time, Y=Y, MPC=as.factor(b)))
}

# Plot
ggplot(sim_data, aes(x=Time, y=Y)) +
  geom_line(color="steelblue") +
  geom_point(color="steelblue") +
  facet_wrap(~MPC, scales="free_y") +
  labs(title="Keynesian Multiplier Simulation for Different MPCs",
       x="Time period", y="Income Yt") +
  theme_minimal()
```